

Generative Interaction Models

Xiaojun Bi

Professor

Department of Computer Science

Stony Brook University



Importance of Generative Interaction Models

- Understanding Human Interaction Beyond Observation
- Scalable Simulation for Design and Evaluation
- Bridging Classical HCI Models and Modern AI



Three Approaches to Generative Interaction Modeling

- Data-driven Generative Behavior Models

Example: WordGesture-GAN, TouchTyping-GAN

- Sequential Decision Behavior Models

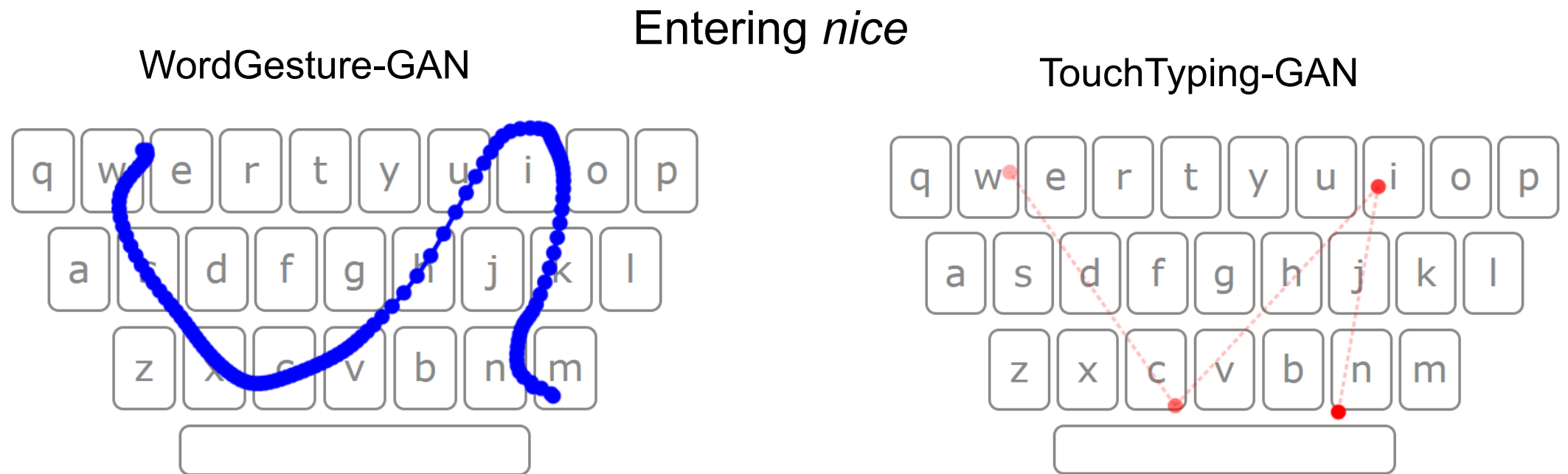
Example: Modeling menu selection as a POMDP

- Bayesian Hierarchical Input Models

Example: Bayesian Hierarchical Pointing Model

Data-driven Generative Behavior Models

- Generating Typing Input using Generative Adversarial Networks



J. Chu, D. An, Y. Ma, W. Cui, S. Zhai, X. Gu, X. Bi "WordGesture-GAN: Modeling Word-Gesture Movement with Generative Adversarial Network ". *CHI 2023*. Article No.: 287 Pages 1–15.

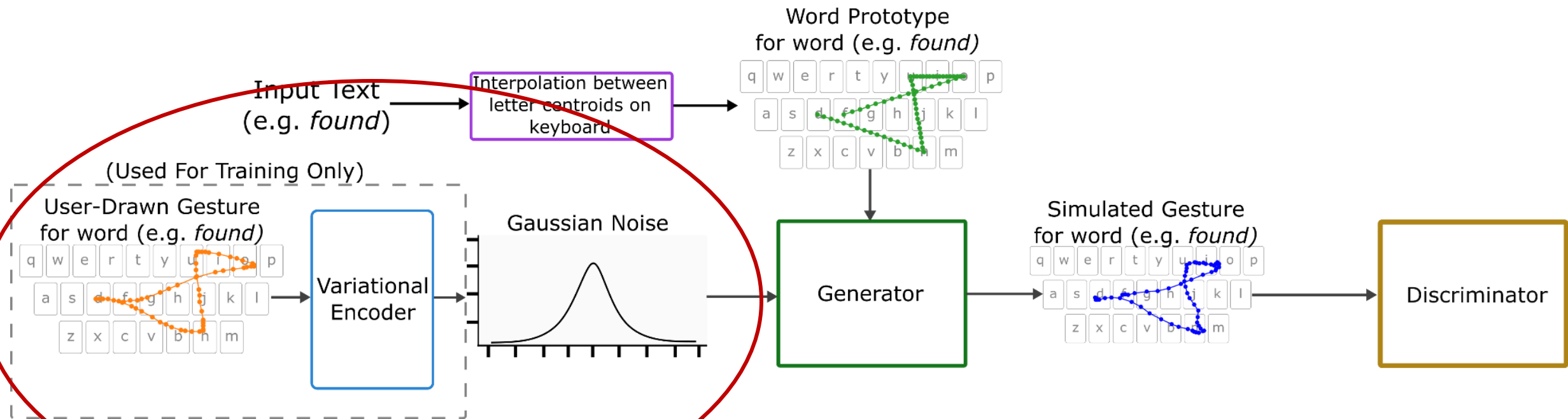
J. Chu, Y. Ma, S. Zhai, X. Gu, X. Bi "TouchTyping-GAN: Modeling Touch Typing with Generative Adversarial Network ". *UIST 2023*. Article 29. Pages 1 - 13

Live Demo

<http://wordgesturegan.com/>

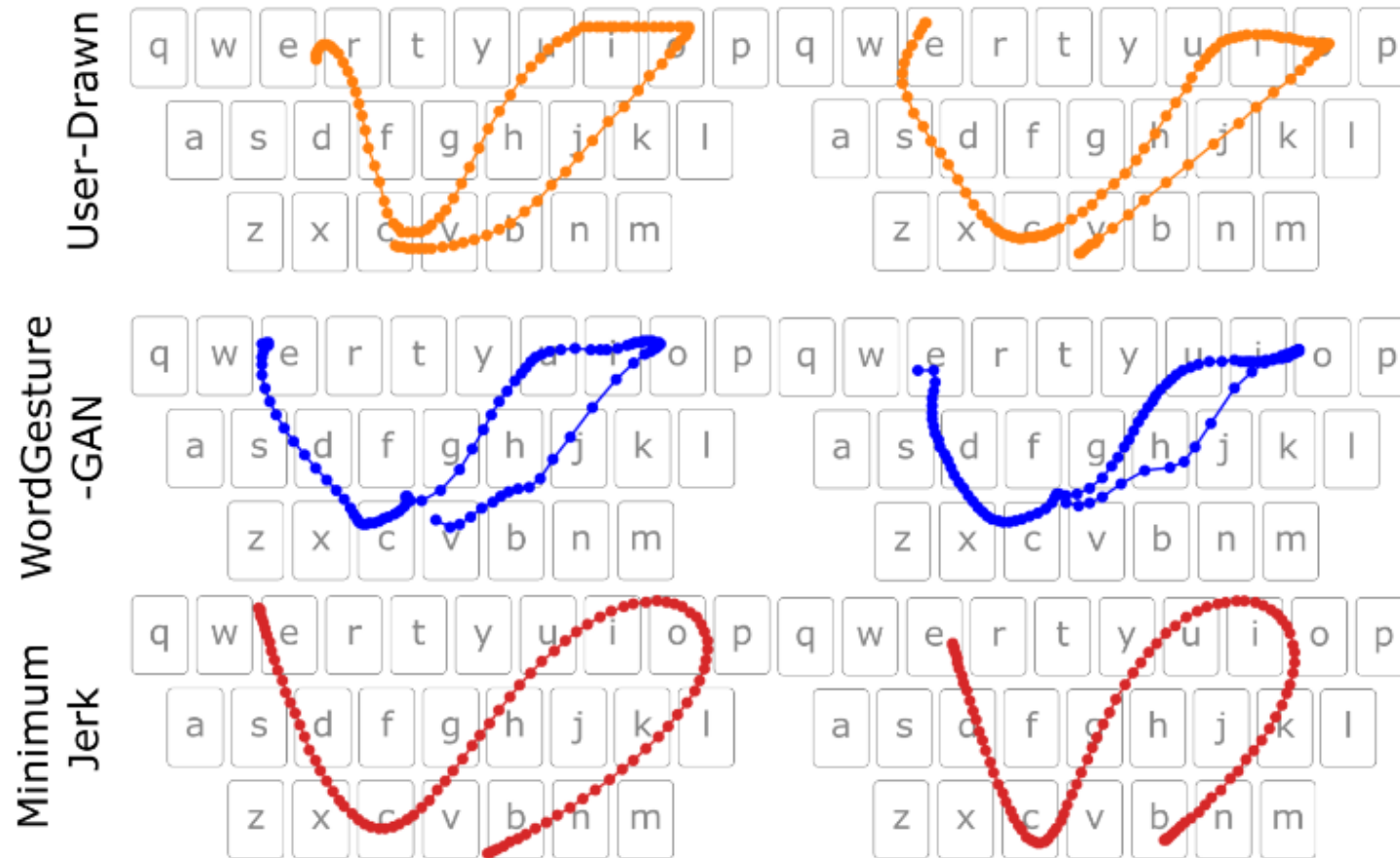
Architecture of WordGesture-GAN

Using a Variational Encoder to control the latent space.



Touch Gestures Generated by Word-Gesture GAN

Sampled Gestures for the Word "voice"

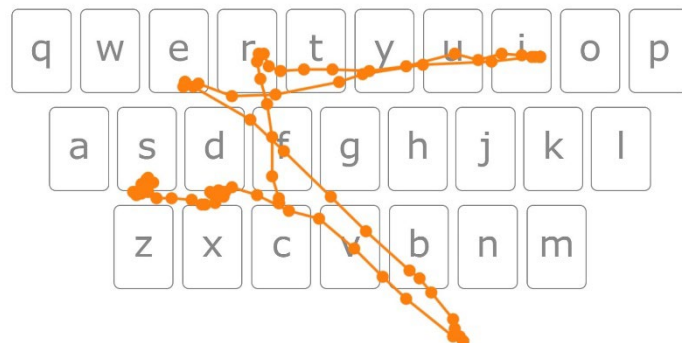


Touch Gestures vs. 3D Gestures



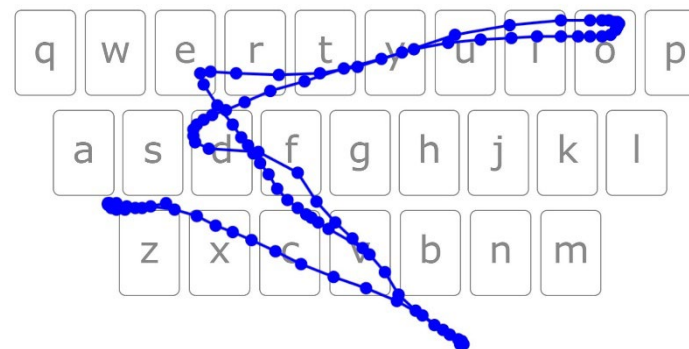
User-drawn 3D Gesture

friends



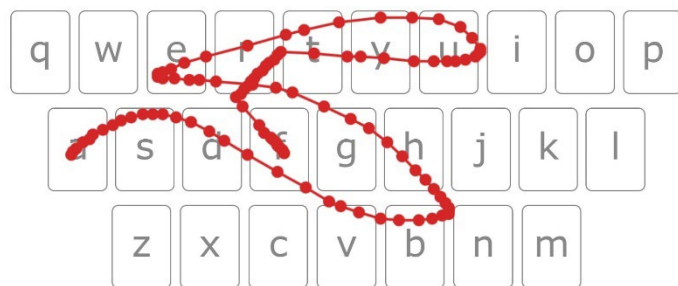
Generated 3D Gesture

friends



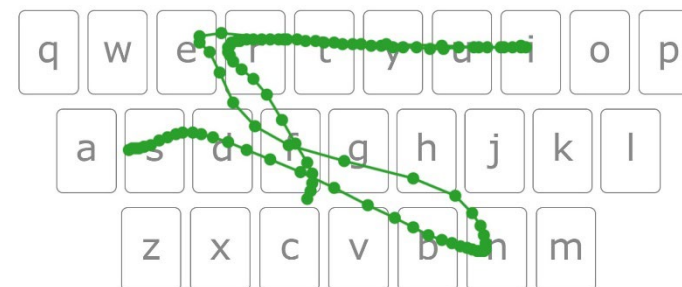
User-drawn Touch Gesture

friends

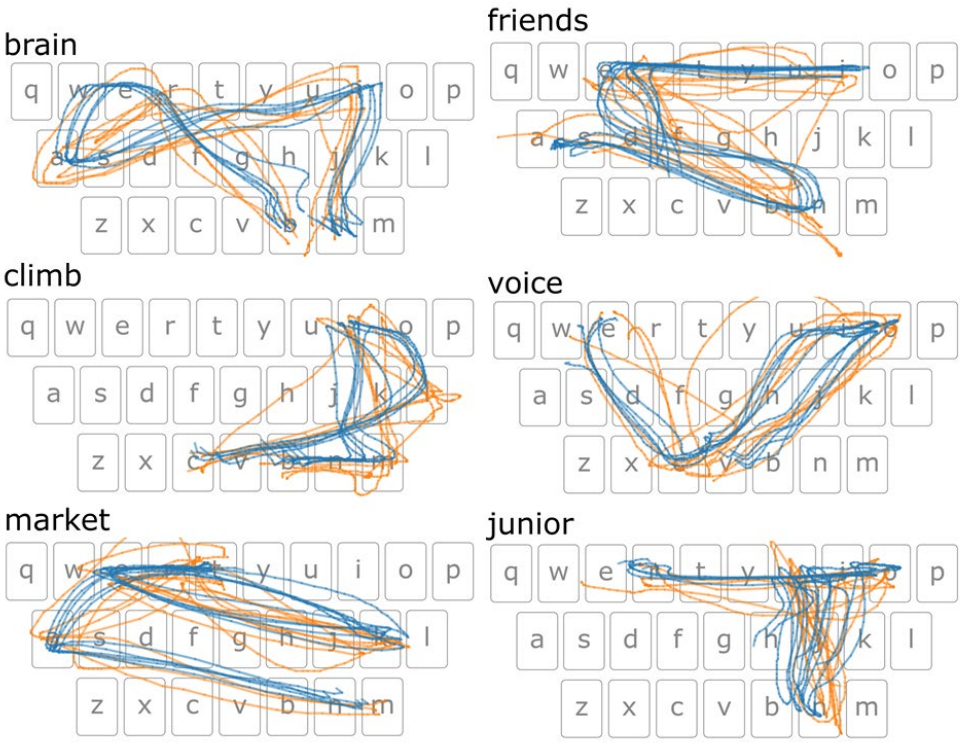


Generated Touch Gesture

friends



User-drawn (Orange) vs. Simulated (Blue)



Training with simulated data improves the accuracy of gesture typing decoder.

Training Setup (# of gestures)	Error Rate After Training
200 user-drawn	32.8%
10000 simulated	28.6%
10000 user-drawn	27.8%

Our GAN models are adopted by



Google's Gboard



Grammarly Keyboard

Three Approaches to Generative Interaction Modeling

- Data-driven Generative Behavior Models

Example: WordGesture-GAN, TouchTyping-GAN

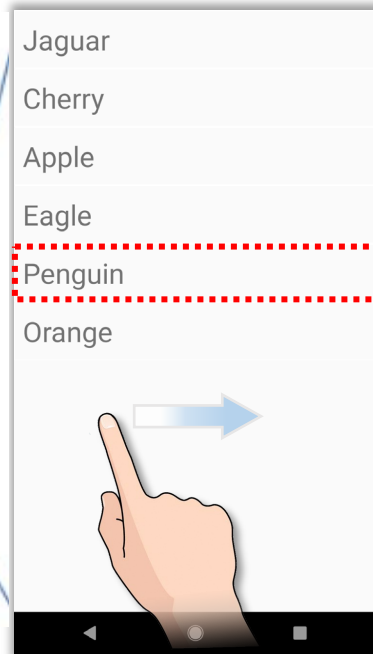
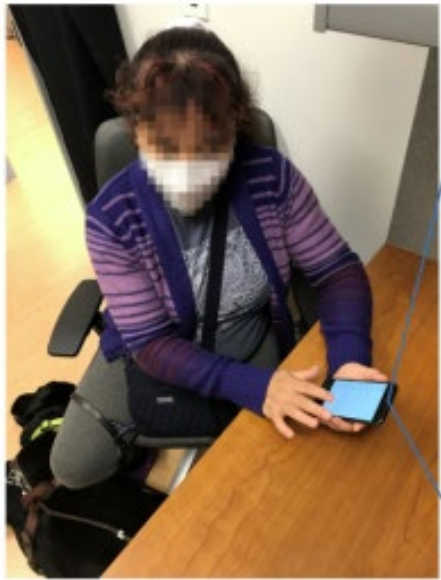
- Sequential Decision Behavior Models

Example: Modeling menu selection as a POMDP

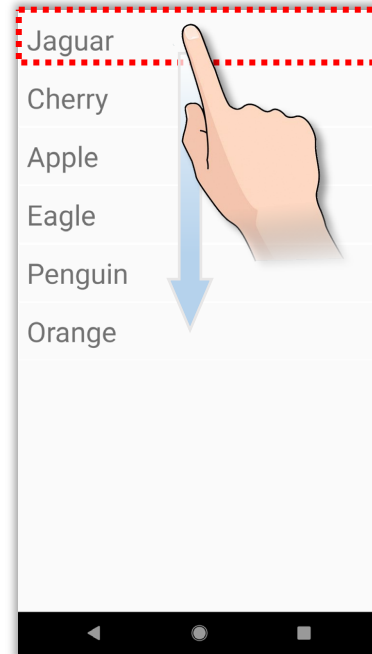
- Bayesian Hierarchical Input Models

Example: Bayesian Hierarchical Pointing Model

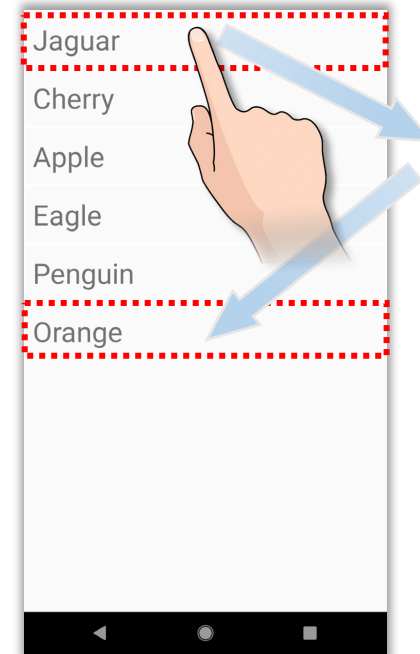
Simulating Menu Selection Behavior of Blind Users



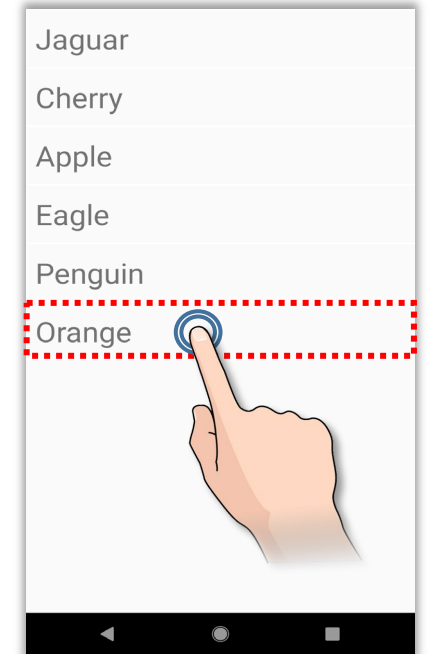
Swiping action



Gliding action



Direct touch action

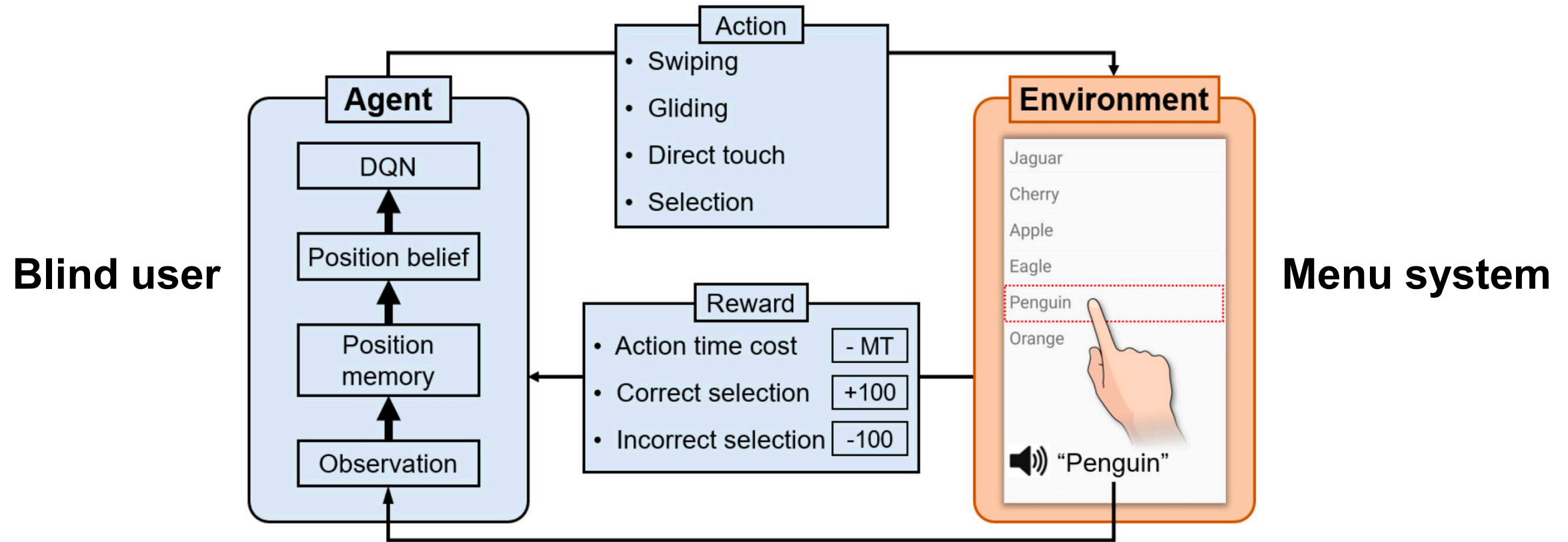


Selection action

Z. Li, Y. Ko, A. Putkonen, S. Feiz, V. Ashok, IV Ramakrishnan, A. Oulasvirta, X. Bi . "Modeling Touch-based Menu Selection for Blind Users". *CHI 2023*. Article No.: 357, pp 1–18



Modeling menu selection as a Partially Observable Markov Decision Process (POMDP)

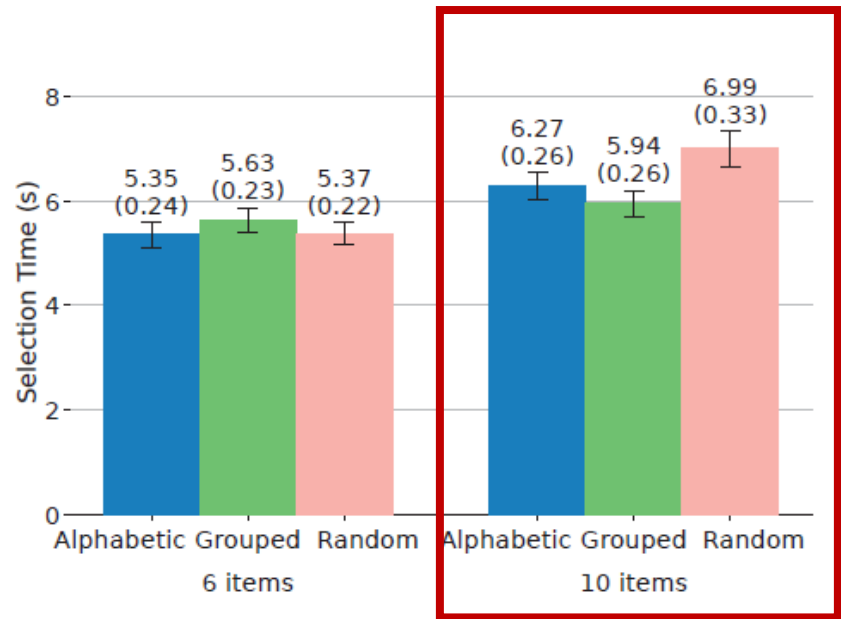




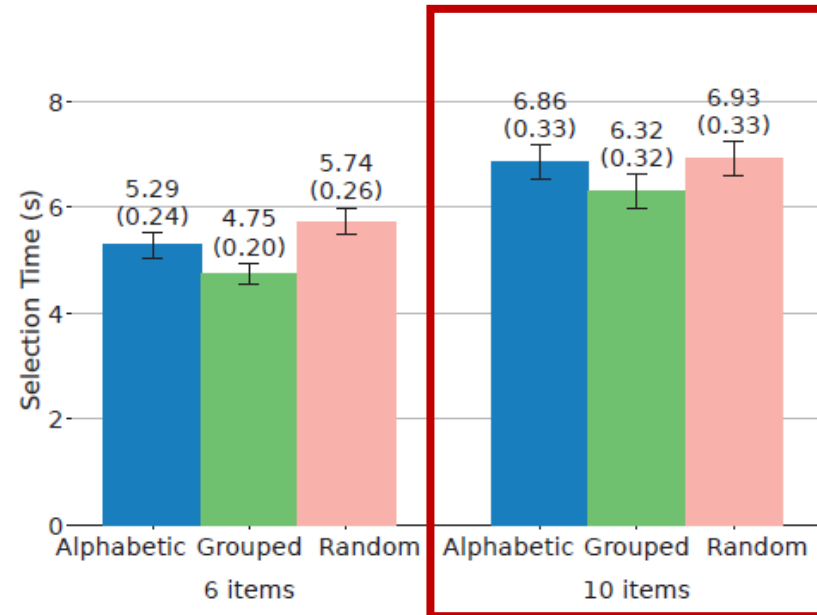
Model Evaluation

Selection time

- MAE 0.39 s (MAPE: 6.71%)



Selection time in the user study



Simulated selection time
by model

Three Approaches to Generative Interaction Modeling

- Data-driven Generative Behavior Models

Example: WordGesture-GAN, TouchTyping-GAN

- Sequential Decision Behavior Models

Example: Modeling menu selection as a POMDP

- Bayesian Hierarchical Input Models

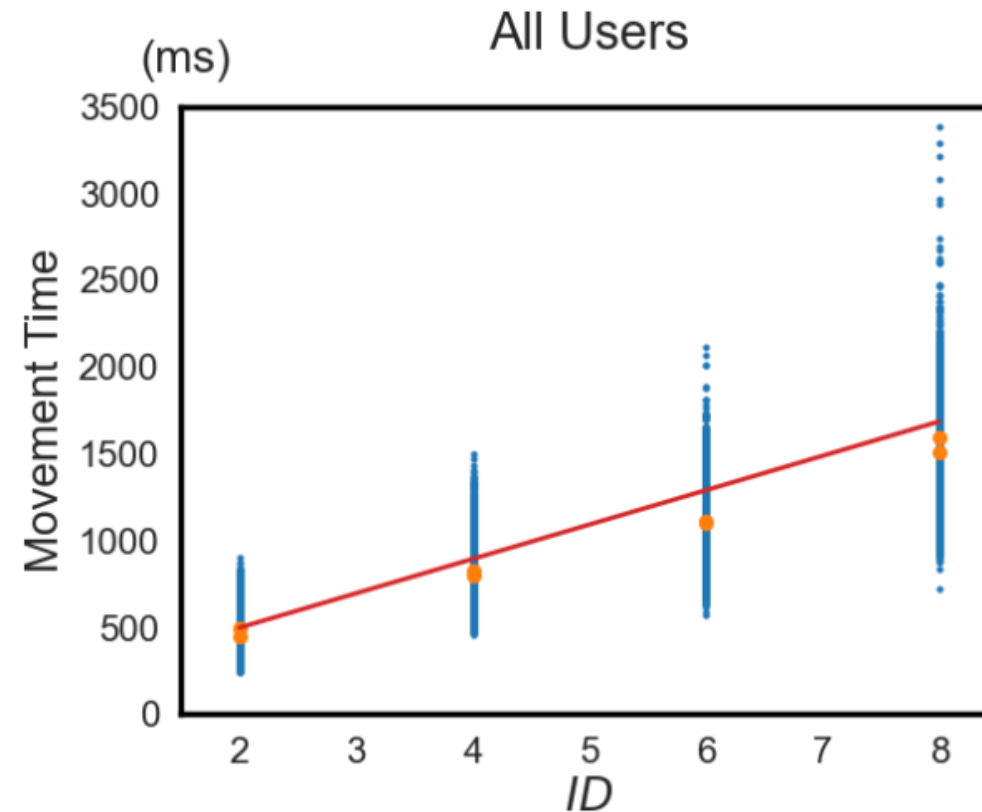
Example: Bayesian Hierarchical Pointing Model

Background

Fitts' Law

$$MT = a + b \cdot \log_2 \left(\frac{A}{W} + 1 \right) [1,2]$$

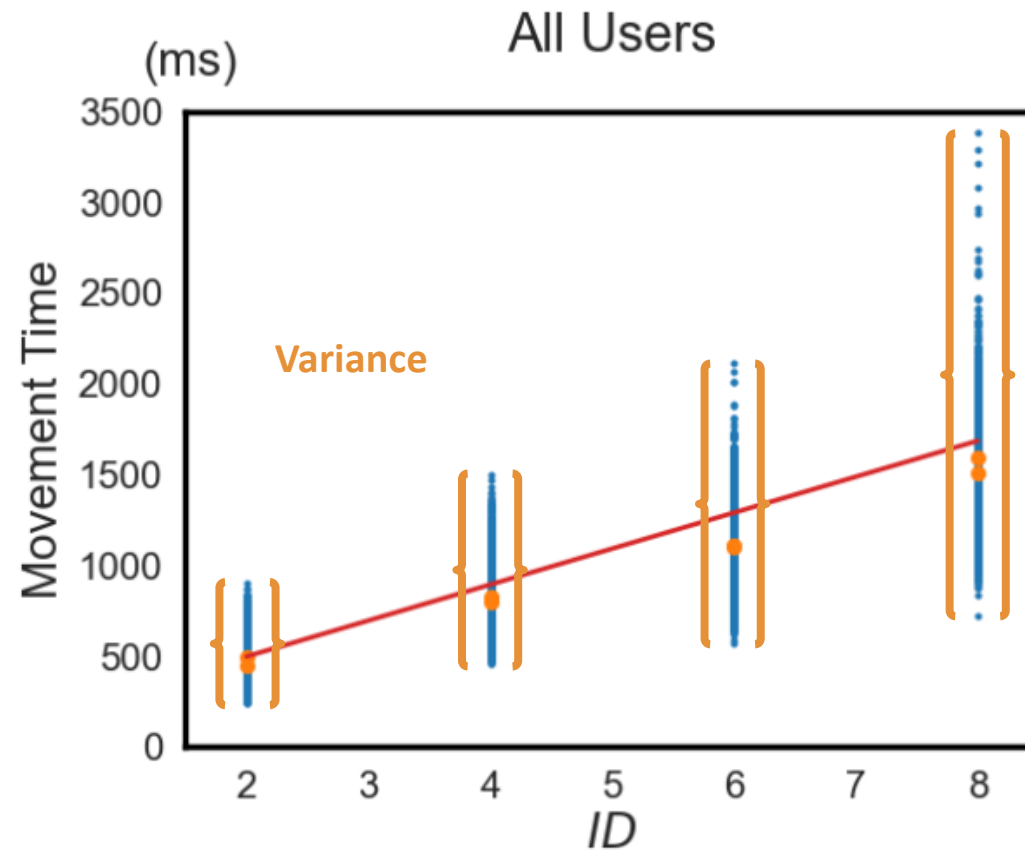
Shortcoming: Predicts the mean of MT only (point estimate)



[1] Paul M Fitts. 1954. The information capacity of the human motor system in controlling the amplitude of movement. Journal of experimental psychology 47, 6 (1954), 381.

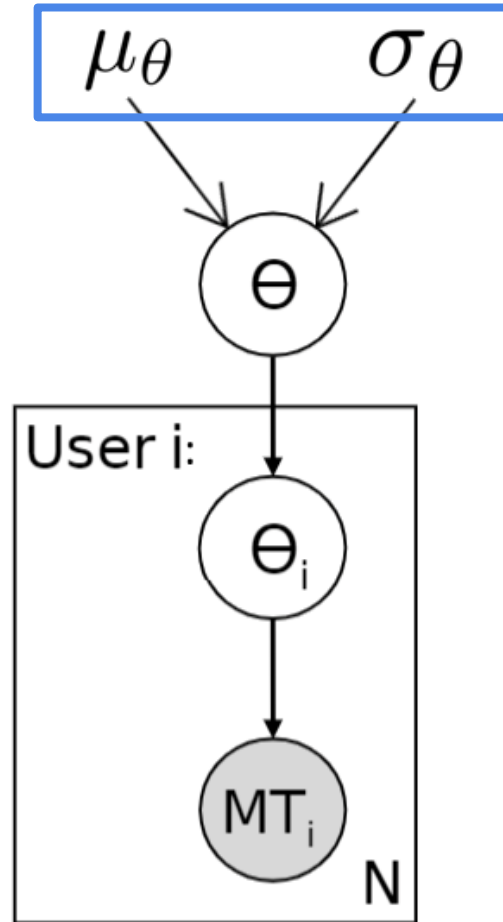
[2] I Scott MacKenzie. 1989. A note on the information-theoretic basis for Fitts' law. Journal of motor behavior 21, 3 (1989), 323–330.

Modeling Variance of Movement Time



$$V : \text{variance}, V = c + d \cdot ID^2$$

Extending Fitts' law to a Bayesian Hierarchical Model



Bayesian Hierarchical Model

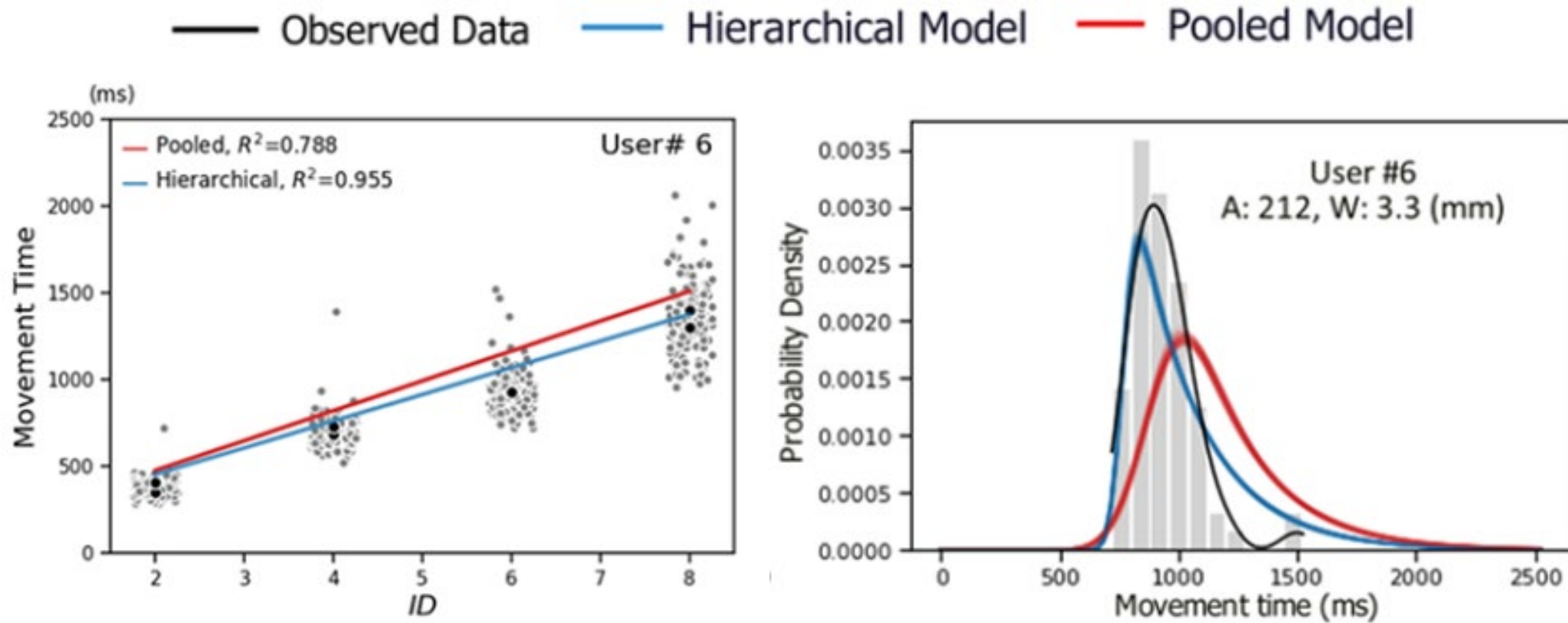
$$\text{Stage-1: } MT_i | \theta_i \sim P(MT_i | \theta_i)$$

We assume that MT_i follows an exGaussian distribution which is governed by the parameter θ_i

$$\text{Stage-2: } \theta_i | \theta \sim P(\theta_i | \theta)$$

It assumes that θ_i follows a distribution governed by the hyperparameter $\theta = (\mu, \sigma^2)$.

Evaluation



The hierarchical (**blue**) model can more accurately predict the distribution of unseen users (**black**) than the pooled (**red**) model.

Three Approaches to Generative Interaction Modeling

- Data-driven Generative Behavior Models

Example: WordGesture-GAN, TouchTyping-GAN

- Sequential Decision Behavior Models

Example: Modeling menu selection as a POMDP

- Bayesian Hierarchical Input Models

Example: Bayesian Hierarchical Pointing Model